Uppsala University



Numerical Linear Algebra

1TD542

Miniproject 1

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September 24, 2026

1 Introduction

One of the main problems of linear algebra is to solve problems of the form

$$Ax = b \tag{1}$$

where $A \in \mathbb{R}^{m \times n}$ is a matrix describing the system, $b \in \mathbb{R}^m$ is a vector of the observed data and x is the unkown solution, like a signal or image. In the special case when A is invertible a unique solution x can be found by computing A^{-1} yielding:

$$x = A^{-1}b$$

However, for the general case with a non-square and overdetermined system (e.g $m > n$) A becomes non-invertible, entailing that the problem lacks a unique solution x. To tackle this issue a common method in finding the best approximation $x_{LS} \approx x$ to the solution is solving the least squares problem:

$$\min_x ||Ax - b||_2^2 \tag{2}$$

In solving the least squares problem one looks for the solution x_{LS} yielding the smallest residual $r = Ax - b$.

The least squares problem is often rewritten to its normal form which then can be solved "as usual":

$$A^T A x_{LS} = A^T b \tag{3}$$

Which often works well if A is full-rank. However, in many practical cases tiny perturbations in the observed data b can cause large changes in the computed solution x_{LS} due to A being ill-conditioned or nearly rank deficient.

To adress this issue in stability, regularisation is commonly used, where a regularisation term is introduced dampening or removing possible singular entries. This project mainly focuses on the popular Tikhonov regularization model:

$$\min_x \underbrace{||Ax - b||_2^2}_{\text{fit-to data term}} + \underbrace{\lambda^2 ||Lx||_2^2}_{\text{regularisation term}} \tag{4}$$

where $\lambda > 0$ is a regularisation parameter and L is a matrix encoding prior smoothness or structural information.

This project discusses this regularisation method in a bit more detail while comparing it to the un-regularised problem and another regularisation method known as a truncated least squares problem.

2 Approach and Method

2.1 Singular Value Decomposition(SVD)

In numerical methods and computational science inverse-computations are generally avoided due to their high computational cost and complexity. Instead, other methods of indirectly finding the inverse are used to solve problems. One such method is called Singular Value Deomposition (SVD), which factorises a problem into blocks that are easier to work with in order to find a solution. This can be done in different ways and the project discusses variations of this method.

In general we SVD decomposition works as follows:

For any matrix $A \in \mathbb{R}^{m\times n}$, we introduce two orthogonal matrices $U \in \mathbb{R}^{m\times m}, \quad V \in \mathbb{R}^{n\times n},$

as well as a diagonal matrix:

$$\Sigma = \begin{bmatrix} \text{diag}(\sigma_1, \ldots, \sigma_r) & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{m\times n},$$

Where the diagonal is built by the singular values

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0,$$

$$\boxed{A = U\Sigma V^T.}$$

When solving the SVD, we use some useful properties which comes from the definitions and properties of the matrices induced:

$$\begin{aligned} A^T &= V\Sigma^{-1}U^T \\ A^TA &= V\Sigma U^TU\Sigma^{-1}V^T \\ U^TU &= I \quad \text{(since U is orthogonal)} \\ \Sigma^{-1}\Sigma = \sigma_i^2 &\text{ and } \Sigma^{-1} = \sigma_i \text{ for } i = 1, ..., n \\ \Rightarrow A^TA &= V\Sigma^{-1}\Sigma V^T \end{aligned}$$

2.2 Approaches to Ill-conditioned problems

As mentioned in the introduction, ill-conditioned problems are commonly adressed using regularisation such as the generalized Tikhonov regularisation (see eq. (4)).

If we assume that A is a full-rank matrix one can show that a normal form representation of the problem of the form

$$(A^TA + \lambda^2 L^T L)x_{RLS} = A^T b \tag{5}$$

can be found. To show this we begin by rewriting the regularised least squares problem to the following form:

$$\min_x ||Ax - b||_2^2 + \lambda^2 ||Lx - 0||_2^2$$

We notice that we can factorise and stack the squared norms in a way that resembles the form of a regular least squares problem (to which we know the normal form of):

$$\begin{aligned} & || \begin{bmatrix} A \\ \lambda^2 L \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} ||_2^2 \\ \implies & \min_x || \underbrace{\begin{bmatrix} A \\ \lambda^2 L \end{bmatrix}}_{M} x - \underbrace{\begin{bmatrix} b \\ 0 \end{bmatrix}}_{d} ||_2^2 \\ \Leftrightarrow & \min_x ||Mx - d||_2^2 \end{aligned}$$

Where we now can rewrite to the normal form of the standard least squares problem as we showed in eq. (3), giving us:

$$M^T M x_{RLS} = M^T d$$

Where x_{RLS} is the regularised least squares solution approximating x. If we write out M and d we find:

$$\begin{aligned} \begin{bmatrix} A^T & \lambda L^T \end{bmatrix} \begin{bmatrix} A \\ \lambda L \end{bmatrix} x_{RLS} &= \begin{bmatrix} A^T & \lambda L^T \end{bmatrix} \begin{bmatrix} b \\ 0 \end{bmatrix} \\ \Leftrightarrow (A^TA + \lambda^2 L^T L)x_{RLS} &= A^T b \end{aligned} \tag{6}$$

If we assume that A is full-rank we can show that a unique regularised least squares solution exists by showing that the left hand side of the problem

(e.g $A^TA+\lambda^2L^TL$) is positive definite. We recall from the lecture notes that a matrix $B \in \mathbb{R}^{n\times n}$ is positive definite if

$$x^TBx > 0, \quad \forall x, \quad x \neq 0 \tag{7}$$

We test this on $B = A^TA + \lambda^2L^TL$

$$\begin{aligned} \implies x^T(A^TA + \lambda^2L^TL)x &= x^TA^TAx + \lambda^2x^TL^TLx \\ &= (Ax)^TAx + \lambda^2(Lx)^TLx \\ &= ||Ax||_2^2 + \lambda^2||Lx||_2^2 \end{aligned}$$

If we assume that the rgularisation parameter $\lambda > 0$ and set $L = I$ we see that the epression is strictly larger than 0 $\quad \forall x \neq 0$;

$$||Ax||_2^2 + \lambda^2||x||_2^2 > 0 \tag{8}$$

Implying that $A^TA + \lambda^2I$ is positive definite and thus yields a unique solution to the regularised least squares problen (see eq.(6)). For a more general L we see that a A and L are required to share the same number of columns in order for a unique solution to exist.

$$||Ax||_2^2 + \lambda^2||Lx||_2^2 > 0$$

2.3 Using SVD to compute the least squares solution

To finally compute the solution to the least squares problem we can use SVD-factorisation in order to compute x_{RLS}. This becomes particularly useful since it avoids the very costly inverse computation otherwise required, as discussed in sec. (2.1). If we factorise A into its SVD factors, e.g $A = U\Sigma V^T$, we note that we have singular values $\sigma > 0, \sigma_r = \sigma_n > 0$ due to A being full-rank. In the case of $L = I$ we can use the factorisation to find our solution regularised solution in the following manner:

$$\begin{aligned} (A^TA + \lambda^2I)x_{RLS} &= A^Tb \\ \Leftrightarrow (V\Sigma^T\Sigma\underbrace{U^TU}_{=I}V^T + \lambda^2\underbrace{I}_{=VV^T})x_{RLS} &= V\Sigma^TU^Tb \\ \Leftrightarrow (V\Sigma^T\Sigma V^T + \lambda^2VV^T)x_{RLS} &= V\Sigma^TU^Tb \end{aligned}$$

$$\Leftrightarrow V(\Sigma^T\Sigma + \lambda^2 I)V^T x_{RLS} = V\Sigma^T U^T b$$
$$\Leftrightarrow (\Sigma^T\Sigma + \lambda^2 I)V^T x_{RLS} = \Sigma^T U^T b$$

We define $y = V^T x_{RLS}$ (meaning that $x_{RLS} = Vy$), giving us:

$$\Leftrightarrow (\Sigma^T\Sigma + \lambda^2)y = \Sigma^T U^T b \tag{9}$$

We remember that Σ is a diagonal matrix with singular values σ_i along its diagonal. As such, we can rewrite the column wise elements of eq. (9) as:

$$(\sigma_i^2 + \lambda^2)y_i = \sigma_i u_i^T b$$
$$\Leftrightarrow y_i = \frac{\sigma_i}{\sigma_i^2 + \lambda^2} u_i^T b \quad , i = 1, ..., n \tag{10}$$

If we now define filter factors of the form

$$\phi_i(\lambda) = \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2} \tag{11}$$

We can write the solution as

$$y_i = \frac{\phi_i(\lambda)}{\sigma_i} u_i^T b \tag{12}$$

Where the filter helps dampen the otherwise problematic instance of very small singular values approaching 0. Meaning, for singualr values $\sigma_i = 0$ we avoid the singular $\frac{1}{0}$ term and instead get $\frac{0}{\lambda^2} = 0$. Which for the filter implies the following relation to the size of λ:

$$\begin{cases} \lim_{\lambda \to 0} \phi(\lambda) = 1 \\ \lim_{\lambda \to \infty} \phi(\lambda) = 0 \end{cases}$$

2.4 Truncated least square solution

As mentioned a problem with the SVD is that some singular values σ_i can be very small if A is ill-conditioned causing the coefficients of $x_{LS} = \sum_{n=1}^{r} \frac{u_i^T b}{\sigma_i} v_i$ to be very large. Another regularisation method adressing this issue, instead of using a Tikhonov regularisation model, is to truncate the singular values σ_i such that $\sigma_1 \geq \sigma_2 \geq ... \geq \sigma_k > \delta$. This truncation creates a new matrix

A_k of rank k such that $A_k = U_k \Sigma_k V_k^T$ with the best rank k approximation to A in the norm:

$$||A - A_k||_2 = \sigma_{k+1} \leq \delta$$

since σ_{k+1} is the largest singular value that has been removed from A. This gives the solution on the form:

$$x_{TLS} = \sum_{i=!}^{k} \frac{u_i^T b}{\sigma_i} v_i$$

Where u_i is the columns of U_k (with k columns) and v_i is the columns if V_k. We can use the same way to interpret a filter as in the previous solution by setting $x_{TLS} = V_k y$. Where the filter in this case removes all the elements where $i > k$. This can easily be done by choosing a filter

$$\phi = \begin{cases} 1 & , i \leq k \\ 0 & , i > k \end{cases}$$

ϕ_i such that $\phi_i = 0$ for $i > k$. If we compare the truncated least squares solution with the Tikhonov solution we can see the following properties:

	Truncated	**Tikhonov**
ϕ_i	1 or 0	$\frac{\sigma_i^2}{\sigma_i^2 + \lambda^2}$
Cutoff	Hard	Soft
Parameter	k, integer	λ, continuous

Table 1: Comparison of truncated SVD and Tikhonov regularisation.

2.5 Generalized SVD

In the prevoius case of solving the regularised least squares problem using SVD, we worked with the simplified system $L = I$ in Tikhonov regularisation. Even though the SVD method works, it can often be useful to use a more stable algorithm such as the generalised SVD method (GSVD). Similarly to SVD, GSVD factorises matrices, however, in pairs instead of one matrix at

a time. For two matrices $A \in \mathbb{R}^{m\times n}$ and $L \in \mathbb{R}^{p\times n}$, we can factorise the matrices using orthogonal matrices $U \in \mathbb{R}^{m\times m}$, $V \in \mathbb{R}^{p\times p}$ and $W \in \mathbb{R}^{n\times n}$ (where W is invertible) such that:

$$A = U\Sigma_A W^{-1}, \quad L = V\Sigma_L W^{-1} \tag{13}$$

under the assumption $m \geq n$ and

$$\operatorname{rank}(M) = r \leq n, \quad M = \begin{bmatrix} A \\ L \end{bmatrix}$$

Where

$$\Sigma_A = \begin{bmatrix} \operatorname{diag}(\alpha_1, \ldots, \alpha_r) & 0 \\ 0 & 0 \end{bmatrix}, \quad \Sigma_L = \begin{bmatrix} \operatorname{diag}(\beta_1, \ldots, \beta_r) & 0 \\ 0 & 0 \end{bmatrix} \tag{14}$$

$$\text{with } 0 \leq \alpha_1 \leq \cdots \leq \alpha_{k+1} = \cdots = \alpha_r = 1,$$
$$\text{and } 1 \geq \beta_1 \geq \cdots \geq \beta_{k+1} = \cdots = \beta_r = 0,$$

where

$$k = \min\{p, r\}.$$

Giving us the useful relation between α and β

$$\alpha_i^2 + \beta_i^2 = 1, \qquad i = 1, 2, \ldots, r.$$

and the ratio

$$\gamma_i = \frac{\alpha_i}{\beta_i}, \qquad i = 1, 2, \ldots, r.$$

γ_i are the generalized singular values. In a similar fashion as done in sec. (2.3) we can use the GSVD to compute the regularised least squares solution (see eq. (5)). To do this we factorise A and L as discussed above, which gives us the equation

$$\implies [(W^{-1})^T \Sigma_A^T \underbrace{U^T U}_{=I} \Sigma_A W^{-1} + \lambda^2((W^{-1})^T \Sigma_L^T \underbrace{V^T V}_{=I} \Sigma_L W^{-1})] x_{RLS}$$

$$= (W^{-1})^T \Sigma_A^T U^T b$$

$$\Leftrightarrow [(W^{-1})^T \Sigma_A^T \Sigma_A W^{-1} + \lambda^2((W^{-1})^T \Sigma_L^T \Sigma_L W^{-1})] x_{RLS} = (W^{-1})^T \Sigma_A^T U^T b$$

$$\Leftrightarrow (W^{-1})^T[\Sigma_A^T\Sigma_A + \lambda^2\Sigma_L^T\Sigma_L]W^{-1}x_{RLS} = (W^{-1})^T\Sigma_A^T U^T b$$

We define $y = W^{-1}x_{RLS}$ (meaning that $x_{RLS} = Wy$)

$$\begin{aligned}
[\Sigma_A^T\Sigma_A + \lambda^2\Sigma_L^T\Sigma_L]y &= \Sigma_A^T U^T b \\
\implies (\alpha_i^2 + \lambda^2\beta_i^2)y_i &= \alpha_i u_i^T b \\
\Leftrightarrow y_i &= \frac{\alpha_i}{\alpha_i^2 + \lambda^2\beta_i^2}u_i^T b
\end{aligned} \tag{15}$$

We notice how we once again can introduce a filter factor $\phi(\lambda)$

$$\phi(\lambda) = \frac{\alpha_i^2}{\alpha_i^2 + \lambda^2\beta_i^2} \tag{16}$$

allowing us to simplify y_i to

$$y_i = \frac{\phi(\lambda)}{\alpha_i}u_i^T b$$

where we once again notice how the filter regularises the solution by removing the singular issue otherwise present when $\alpha_i \to 0$.

2.6 Signal processing and deblurring

To exemplify the above discussed theory we use an example within signal processing and deblurring. When measuring a singal, the measurement is almost never the true signal due to noise in the measurement data. An issue commonly adressed using deblurring. When deblurring, we use an integral equation of the form:

$$\int K(s,t)x(t)\,dt + e(s) = b(s) \tag{17}$$

Where $s, t \in \mathbb{R}^d$ represent spatial information, $K(s,t)$ is the kernel (also called the point spread function which defines the blur), $x(t)$ is the true signal and $e(s)$ is the error. If we discretise the function, we get a linear system of the form $b = Ax + e$, where A now is the blurring operator. This becomes a least squares problem that can then be solved using GSVD. To show how this can be done we look at a one-dimensional problem with a true signal

$$x(t) = e^t \sin(\pi t), \qquad t \in [-1, 1] \tag{18}$$

We then discretize and approximate the integral equation (see eq. (17)) using the box quadrature rule at the sampling points t_i, with the gaussian kernel

$$K(s,t) = c\exp\left(-(s-t)^T\beta(x-t)\right), \qquad \beta, c > 0. \tag{19}$$

For the measurement error, we define $e = \delta\mathcal{N}(0,1)$ and use the first derivative finite-difference approximation L to solve the problem using GSVD.

3 Results

When doing the signal processing and deblurring we can plot the regularised and unregularised least-squares solution as well as the true signal, the blurred noisy data and the reconstructed signals to compare the quality of the reconstructions.

When using a noise amplitude $\delta = 10^{-1}$, $\beta = 100$, $c = 1$, regularisation parameter $\lambda = 0.5$ on a $\mathbb{R}^{20\times 20}$ system, we see the following relations:

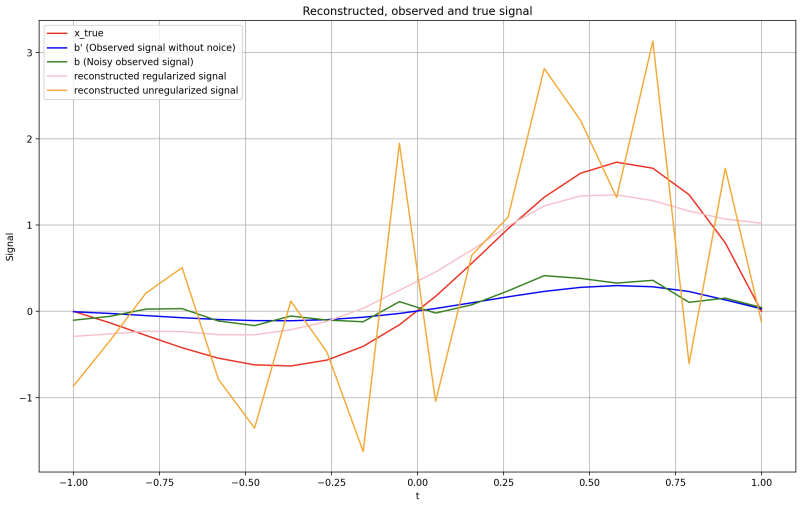


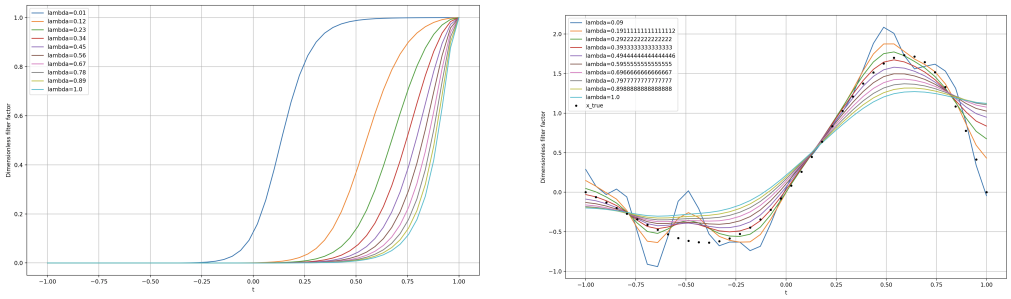
Figure 1: Plots of reconstructed, observed and true signal.

From the figure, we can clearly see that the regularised reconstructed signal creates a much better and closer approximation of the true signal when compared to the unregularised reconstructed signal.

As discussed, the choice of regularisation parameter λ affects how much

of the signal is let through the general filter factor $\phi(\lambda)$. To visualise how the filter factor changes depending on the regularisation parameter, we plotted

To find the best value of λ for the filter while deblurring, we plotted different $\phi(\lambda)$ with fixed value of λ ove rthe same discrete interval t_n, and varied the value on different plots, see figure 2a



(a) Filter factors $\phi(\lambda)$ for different values of λ.

(b) Signal reconstruction for different values of $\lambda \in [0.09, 1.0]$.

Figure 2: Effect of varying the regularization parameter λ.

This shows how, for a higher λ, more of the singular values are suppressed by the filter factors.

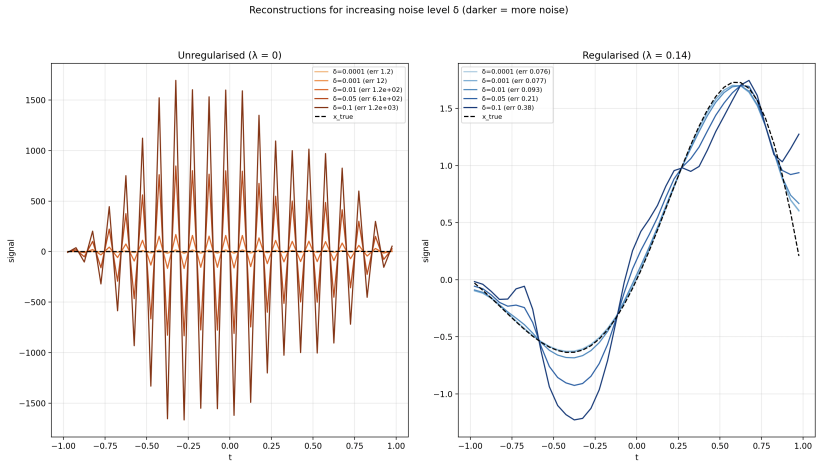


Figure 3: Unregularised (left) and regularised (right) reconstructed signal with varying noise-amplitudes

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4 Discussion

4.1 Choosing the best λ for the filter factor

When analysing fig (2a), we can clearly see that the filter cancels out more and more of the singular values for bigger λ.This is useful since, for the generalized SVD, the smallest singular values are stored as the first σ_i. In fact, for the first σ_i with $i = 1, ..., 30$, the singular values are zero. Upon inspection of fig (2b), the choice of λ has a significant effect on the reconstructed signal, and finding a good value while compromizing between noice and over-fitting, using fig (2b), would be be a smaller λ closer to zero but $\lambda > 0$.

The under-regularized signal is more "noisy", varying and with a larger amplitude. In opposite, the over-regularized sginals are smoothened out, but not reaching the same amplitued, and therefore losing part of the reconstruction, hence being over-fitted.

4.2 Finding the best suited λ using the L-curve

Since the regularization problem with a choice of the parameter λ is a compromise between over-fitting the regularization data, and also having the noise. One way to theoretically face this problem is the so called L-curve. This is executed by simply computing the logarithms of the norm of the fit-to-data term with varying λ_i, and compare them to the norm of the regularization term $||Lx_\lambda||$, as done in the following following graph. Note that λ is a parameter along the curve.

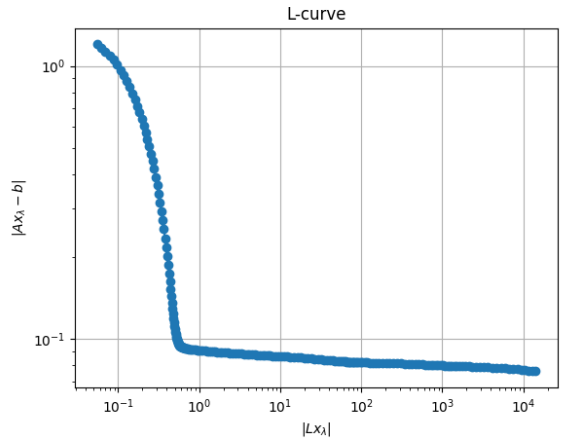


Figure 4: The L-curve with λ varying with logspace(-8,-1,200)

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The horizontal part of the curve corresponds to under-estimates (λ being too small), and the data is sensitive for the noise. The opposite is seen for the flat part of the curve, with over-smoothing as the problem. The point in the "corner", where the curve flattens out sharply, is therefore seen as a good approximation and compromise between these two. Finding this value gives us a $\lambda \simeq 0.14$. This was found by finding the point with maximun curvature and using that index to find which lambda in the logspace(-6,1,200).

4.3 How noise affects the solution

When changing the amplitude of the noice one quickly sees how the unregularised solution returns a much more accurate reconstruction of the true signal, as can be seen in fig. (3. Which aligns with the expected result since an increase in noice reduces the size of the singular values of the system creating the singular terms discussed in sec. (2.2).

5 References

[1] D. Mirzaei, Numerical Linear Algebra, Lecture notes, Uppsala University, 2023

[2] W. C. Karl, "Regularization in Image Restoration and Reconstruction," i Handbook of Image and Video Processing, A. C. Bovik, red., 2:a uppl. San Diego, CA, USA: Academic Press, 2005.